

Artificial Intelligence using Python

Hands-on Exercise No. 3 DigiSkills 3.0 Batch-03

Hands-on Exercise No. 3

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Problem Statement

House_Price_Prediction

Task 1:

Train-Test Split:

- Split the dataset into training and testing sets
- Use an appropriate ratio (e.g., 80% training, 20% testing)

Solution:

```

1 import numpy as np
2 import pandas as pd
3 from sklearn.model_selection import train_test_split
4
5 df = pd.read_csv("HousePricePrediction.csv")
6
7 df = df.dropna()
8
9 print(df.head(10))
10
11 X = df.drop("SalePrice", axis=1)
12 y = df["SalePrice"]
13
14 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
15 random_state=42)
16
17 print("Training features shape:", X_train.shape)
18 print("Testing features shape:", X_test.shape)
19 print("Training target shape:", y_train.shape)
20 print("Testing target shape:", y_test.shape)

```

	Id	MSSubClass	MSZoning	LotArea	LotConfig	BldgType	OverallCond	\
0	0	60	RL	8450	Inside	1Fam	5	
1	1	20	RL	9600	FR2	1Fam	8	
2	2	60	RL	11250	Inside	1Fam	5	
3	3	70	RL	9550	Corner	1Fam	5	
4	4	60	RL	14260	FR2	1Fam	5	
5	5	50	RL	14115	Inside	1Fam	5	
6	6	20	RL	10084	Inside	1Fam	5	
7	7	60	RL	10382	Corner	1Fam	6	
8	8	50	RM	6120	Inside	1Fam	5	
9	9	190	RL	7420	Corner	2fmCon	6	

	YearBuilt	YearRemodAdd	Exterior1st	BsmtFinSF2	TotalBsmtSF	SalePrice
0	2003	2003	VinylSd	0.0	856.0	208500.0
1	1976	1976	MetlSd	0.0	1262.0	181500.0
2	2001	2002	VinylSd	0.0	920.0	223500.0
3	1915	1970	Wd Sdng	0.0	756.0	140000.0
4	2000	2000	VinylSd	0.0	1145.0	250000.0
5	1993	1995	VinylSd	0.0	796.0	143000.0
6	2004	2005	VinylSd	0.0	1686.0	307000.0
7	1973	1973	HdBoard	32.0	1107.0	200000.0
8	1931	1950	BrkFace	0.0	952.0	129900.0
9	1939	1950	MetlSd	0.0	991.0	118000.0

Training features shape: (1168, 12)
 Testing features shape: (292, 12)
 Training target shape: (1168,)
 Testing target shape: (292,)

Task 2:

Model Training:

- Train a Linear Regression model using the training data.

Solution:

```

1 from sklearn.linear_model import LinearRegression
2 from sklearn.metrics import mean_squared_error, r2_score
3 import pandas as pd
4
5 categorical_cols_train = X_train.select_dtypes(include=['object']).columns
6
7 X_train_processed = pd.get_dummies(X_train, columns=categorical_cols_train,
8 drop_first=True)
9 X_test_processed = pd.get_dummies(X_test, columns=categorical_cols_train,
10 drop_first=True)
11
12 train_cols = X_train_processed.columns
13 test_cols = X_test_processed.columns
14
15 missing_in_test = set(train_cols) - set(test_cols)
16 for c in missing_in_test:
17     X_test_processed[c] = 0
18
19 missing_in_train = set(test_cols) - set(train_cols)
20 for c in missing_in_train:
21     X_train_processed[c] = 0
22
23 X_test_processed = X_test_processed[train_cols]
24
25 model = LinearRegression()
26 model.fit(X_train_processed, y_train)
27
28 y_pred = model.predict(X_test_processed)
29
30 mse = mean_squared_error(y_test, y_pred)
31 r2 = r2_score(y_test, y_pred)

```

Task 3:

Model Prediction:

- Use the trained model to predict house prices on the test data

Solution:

```

1 y_test_pred = model.predict(X_test_processed)
2
3 prediction_results = pd.DataFrame({
4     "Actual Price": y_test.values,
5     "Predicted Price": y_test_pred
6 })
7
8 print(prediction_results.head(10))

```

	Actual Price	Predicted Price
0	154500.0	168733.001891
1	325000.0	243223.761211
2	115000.0	114785.412278
3	159000.0	157880.219083
4	315500.0	254517.600860
5	75500.0	156703.578885
6	311500.0	178379.561667
7	146000.0	127064.111212
8	84500.0	133197.418859
9	135500.0	149331.190822

Task 4:

Model Evaluation; Evaluate the model using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

- R^2 Score

Solution:

```

1  from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
2  from sklearn.metrics import mean_absolute_percentage_error
3  import numpy as np
4
5  y_pred = model.predict(X_test_processed)
6
7  print("R2 Score:", r2_score(y_test, y_pred))
8  print("MAE:", mean_absolute_error(y_test, y_pred))
9  print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
10 print("MAPE:", mean_absolute_percentage_error(y_test, y_pred))

```

R2 Score: 0.6196387511449829
MAE: 34129.48682987305
RMSE: 54013.8394792581
MAPE: 0.1978186026868944

Task 5:

Interpretation:

- Analyze the performance of the model.
- Discuss whether the model is accurate or needs improvement.

Solution:

```

1  from sklearn.preprocessing import StandardScaler
2  from sklearn.linear_model import LinearRegression
3  from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
4  mean_absolute_percentage_error
5  import numpy as np
6
7  scaler = StandardScaler()
8  X_train_scaled = scaler.fit_transform(X_train_processed)
9  X_test_scaled = scaler.transform(X_test_processed)
10
11 model = LinearRegression()
12 model.fit(X_train_scaled, y_train)
13
14 y_pred = model.predict(X_test_scaled)
15
16 print("R2 Score:", r2_score(y_test, y_pred))
17 print("MAE:", mean_absolute_error(y_test, y_pred))
18 print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
19 print(mean_absolute_percentage_error(y_test, y_pred))

```

R2 Score: 0.6196387511449681
MAE: 34129.4868298736
RMSE: 54013.83947925916
0.19781860268690482